

MIKE TAYLOR III

Generative AI Engineer II

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PROFESSIONAL SUMMARY

Generative AI Engineer II at Cook Medical building enterprise LLM, RAG, and agentic AI systems for a regulated medical-device environment. Lead developer on CookGPT, the company's internal AI platform, and architect of the Enterprise AI Evaluation Framework covering hallucination risk, retrieval quality, agent reliability, guardrail behavior, structured output compliance, and responsible AI behavior across roughly 750 to 800 test cases and 15 evaluation buckets with weekly regression testing. Work spans PHI-sensitive deployment, prompt-injection resistance testing, EU AI Act and GDPR alignment, and production readiness criteria for agent-based workflows. Currently completing an MBA in Technology Management and beginning doctoral research in AI and machine learning, with focus on agentic architectures, evaluation infrastructure at scale, and governance frameworks for production AI.

CORE EXPERTISE

- **Enterprise LLM Deployment:** production LLM systems, prompt engineering, fine-tuning, inference optimization
- **Agentic AI Systems:** multi-step reasoning, tool use, ReAct/CoT, AutoGen, CrewAI, LangChain, multi-agent orchestration
- **Retrieval-Augmented Generation:** vector search, embeddings, semantic chunking, hybrid retrieval, re-ranking, grounding
- **LLM Evaluation Infrastructure:** hallucination detection, retrieval grounding measurement, regression pipelines, structured output validation
- **Responsible AI & Governance:** guardrails, content safety, policy enforcement, EU AI Act, GDPR, PHI handling, audit logging
- **Production Readiness:** MLOps, observability, monitoring, latency optimization, CI/CD for AI systems

WORK EXPERIENCE

Cook Medical – Generative AI Engineer II

Dec 2025 – Present

Bloomington, IN (Remote)

- Design, build, evaluate, and deploy enterprise LLM, RAG, and intelligent agent systems supporting internal workflows, decision support, and organizational knowledge access across a global medical-device company.
- Serve as one of the lead developers on CookGPT, the company's internal AI platform powered by large language models, retrieval-augmented generation, and agentic orchestration.

- Architect and maintain the Enterprise AI Evaluation Framework covering eight production-readiness dimensions including hallucination risk, retrieval quality, agent reliability, guardrail behavior, structured output compliance, responsible AI behavior, and operational safety, with approximately 750 to 800 test cases organized across 15 evaluation buckets.
- Run weekly regression testing pipelines and hallucination tracking at both temperature 0 and production temperature settings to monitor drift, surface failure modes, and inform go/no-go release decisions.
- Implement prompt-injection resistance testing, PHI protection validation, and structured output compliance checks for regulated, PHI-sensitive deployment environments.
- Drive AI governance practices aligned with EU AI Act and GDPR considerations, including audit logging, traceability, content safety, and policy enforcement for production AI releases.
- Lead development of an AI-powered translation platform with a human-in-the-loop validation pipeline that scores translation quality against approved source materials and keeps professional translators in the review loop for auditability and compliance.
- Lead development of an agentic sustainability questionnaire platform that autonomously completes and routes legally binding compliance documents to target organizations with full traceability.
- Mentor colleagues on AI engineering practices and train company leadership on the effective and responsible use of enterprise AI through hands-on workshops.

All-American School of English – Founder & Technical Lead

Jan 2020 – Dec 2025

Warsaw, Poland

- Founded and operated an online education platform serving software developers and IT professionals at large enterprises across Poland and Europe, delivering specialized technical English instruction tailored to programming and engineering teams.
- Built and maintained the platform infrastructure using JavaScript, React, Node.js, REST APIs, and databases, supporting scheduling, communication workflows, and student management.
- Implemented analytics to track learner progress, engagement, and retention, using the data to adjust curriculum and delivery methods and improve student retention and recurring revenue.
- Managed client relationships, operations, curriculum design, and platform improvements across an international client base.

European League of Football – Data Analyst & Professional Athlete

Jan 2020 – 2024

Berlin, Germany

- Analyzed game film and player performance data using statistical methods to identify opponent formations, play tendencies, and trends, delivering actionable insights to coaching staff.
- Improved defensive read accuracy by approximately 15 percent through data-informed game planning, contributing to stronger in-game adjustments.
- Competed as a professional athlete while applying the same analytical discipline to performance preparation and review.

Georgetown Alumni Funds – Data Analyst & Student Caller

Aug 2017 – Dec 2019

Washington, D.C.

- Tracked and updated daily fundraising results in Excel, maintaining accurate records across alumni donation campaigns.
- Managed high-volume inbound and outbound alumni calls, strengthening donor engagement and contributing to annual fundraising targets.

PROJECTS

CookGPT – Internal Enterprise AI Platform

Cook Medical • Production

- Lead developer on Cook Medical's internal enterprise AI platform built on large language models, retrieval-augmented generation, and intelligent agent orchestration.
- Enables employees across engineering, sustainability, and compliance functions to access institutional knowledge, automate workflows, and receive grounded decision support at enterprise scale.
- Designed for reliability and auditability in a regulated medical-device environment with PHI handling considerations.

Enterprise AI Evaluation Framework

Cook Medical • Production

- Architected and maintain a systematic evaluation framework assessing enterprise AI systems across eight production-readiness dimensions before deployment.
- Approximately 750 to 800 test cases organized into 15 evaluation buckets covering hallucination risk, retrieval quality, agent reliability, guardrail behavior, structured output compliance, responsible AI behavior, and operational safety.
- Implements weekly regression testing, hallucination tracking at temperature 0 and production settings, prompt-injection resistance testing, and PHI protection validation.
- Generates quantitative scores, qualitative risk reports, and remediation guidance that support data-driven go/no-go decisions for production AI releases.

AI Sustainability Questionnaire Agent

Cook Medical • In Development

- Leading development of an agentic platform where an intelligent agent autonomously completes and routes sustainability questionnaires to target companies.
- Executes a structured workflow that populates legally binding compliance documents and submits them to appropriate organizations with full traceability and auditability.
- Built on retrieval-augmented generation against approved source materials and an evaluation layer that validates output structure and completeness before submission.

AI-Powered Translation Platform

Cook Medical • In Development

- Leading development of an AI translation application backed by approved translation documents and a network of professional translators.
- Multi-layer validation pipeline scores translation quality, enforces consistency against approved source materials, and routes outputs to human translators for review at every stage.
- Designed for accuracy, auditability, and regulatory compliance in regulated commercial contexts.

AI-Powered Deep Research Agent

Personal Project • 2024

- Engineered a multi-agent research pipeline that autonomously planned searches, executed queries, summarized findings, and produced structured Markdown reports with iterative critique and revision cycles.
- Deployed with a Gradio UI that guided users through clarifying questions and research refinement, and automated report delivery through the Resend API.

Multi-Agent Software Engineering Crew

Personal Project • 2024

- Built a multi-agent software engineering crew using CrewAI with engineering lead, backend, frontend, and test engineer agents collaborating to plan, develop, test, and document code.
- Configured modular agents.yaml and tasks.yaml definitions for role-specific behaviors and integrated secure Docker-based sandboxes for code testing.

Sickle Cell Image Classification

Personal Project • 2023

- Built and trained a Convolutional Neural Network using TensorFlow and Keras on a curated medical image dataset from Uganda for sickle cell classification.
- Reached 1.00 training accuracy and 0.96 validation accuracy over 30 epochs and outlined data augmentation, dropout tuning, and early stopping as next steps to reduce overfitting.

TECHNICAL SKILLS

AI & LLMs

GPT-4, Claude, Gemini, prompt engineering, fine-tuning, RAG systems, LangChain, LlamaIndex, Semantic Kernel, embeddings, vector databases, LLM evaluation, hallucination detection, structured output validation

Agentic AI

ReAct, CoT, tool use, multi-agent systems, AutoGen, CrewAI, agent orchestration, memory systems, planning agents, agent routing reliability

Machine Learning

PyTorch, TensorFlow, Scikit-learn, Hugging Face, deep learning, NLP, computer vision, model evaluation, ONNX

Data & Engineering

Python, SQL, Pandas, Polars, Apache Spark, dbt, Airflow, ETL pipelines, data modeling, REST APIs, FastAPI, Pydantic

Cloud, MLOps & Platforms

Palantir Foundry, AIP Evals, Azure, AWS, GCP, Azure OpenAI, Docker, Kubernetes, CI/CD, MLflow, Weights & Biases

Responsible AI & Governance

AI governance, EU AI Act, GDPR, PHI handling, bias detection, fairness metrics, explainability, safety alignment, prompt-injection resistance, audit logging

Programming

Python, TypeScript, JavaScript, Node.js, React, C++, SQL, Bash, R

EDUCATION

Walsh College – PhD, Technology – AI & Machine Learning

In Progress

Troy, MI

Doctoral research focused on enterprise AI deployment, evaluation methodology, and the organizational dimensions of AI transformation in regulated environments.

Western Governors University – MBA, Technology Management

In Progress

Online

Technology-focused MBA integrating business strategy, IT management, and organizational leadership.

University of Phoenix – MS, Data Science

Completed

Phoenix, AZ

University of Phoenix – BS, Computer Science

Completed

Phoenix, AZ

Georgetown University – BA, History

Aug 2015 – Dec 2019

Washington, D.C.

NCAA Division I football student-athlete. Coursework emphasized research, analytical reasoning, and evidence-based written communication.

CERTIFICATIONS & ONGOING TRAINING

- **AI in Production: GenAI & Agentic AI at Scale (2025)** – Ed Donner, Ligency – *In Progress*
- **The AI Engineer Course (2025)** – 365 Careers – *In Progress*
- **Azure Data Engineering Masters: Build Scalable Solutions (2025)** – Satyajit Pattnaik & Praveen Kumar – *In Progress*
- **PyTorch for Deep Learning Bootcamp (2025)** – ZTM, Andrei Neagoie & Daniel Bourke – *Completed*
- **Machine Learning, Data Science & AI Bootcamp (2025)** – ZTM – *Completed*

ADDITIONAL

- Languages: English (native), Polish (conversational), German (basic).
- Former NCAA Division I football player at Georgetown University and professional player in the European League of Football.
- Interests: AI evaluation infrastructure, agentic enterprise architectures, ontology-driven AI systems, physics and cosmology, NASA and space exploration, strength training and powerlifting science.